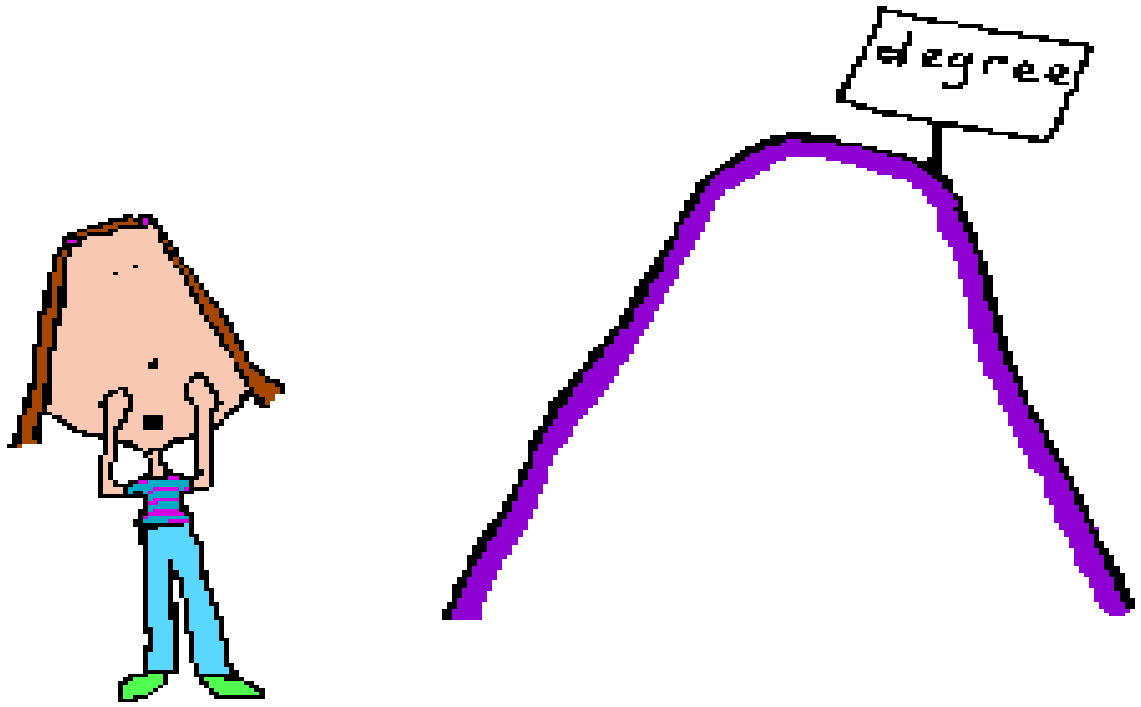
158.337
Database Development



158.337

Learning outcomes

Students who successfully complete this course should be able to:

1. Demonstrate comprehension of database theory and architecture by relating them to real world data applications.
2. Create and modify data and database objects to store, manipulate and analyse enterprise-wide data.
3. Design and implement an application database for a given scenario including the impact of normalization and indexes.
4. Use different programming tools to conduct analytics of stored data.
5. Compare and contrast various data handling, transactional and analytical processing techniques being used for relational and non-relational databases.

158.337

Assessments

Assessment	Learning outcomes assessed	Weighting
Computer programmes	1 2 3 4 5	60%
Exam (centrally scheduled)	1 2 3 4 5	40%

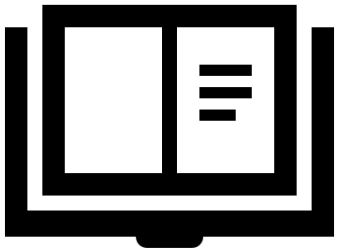
Assessment weightings can change up to the start of the semester the course is delivered in.

You may need to take more assessments depending on where, how, and when you choose to take this course.

Final Exam

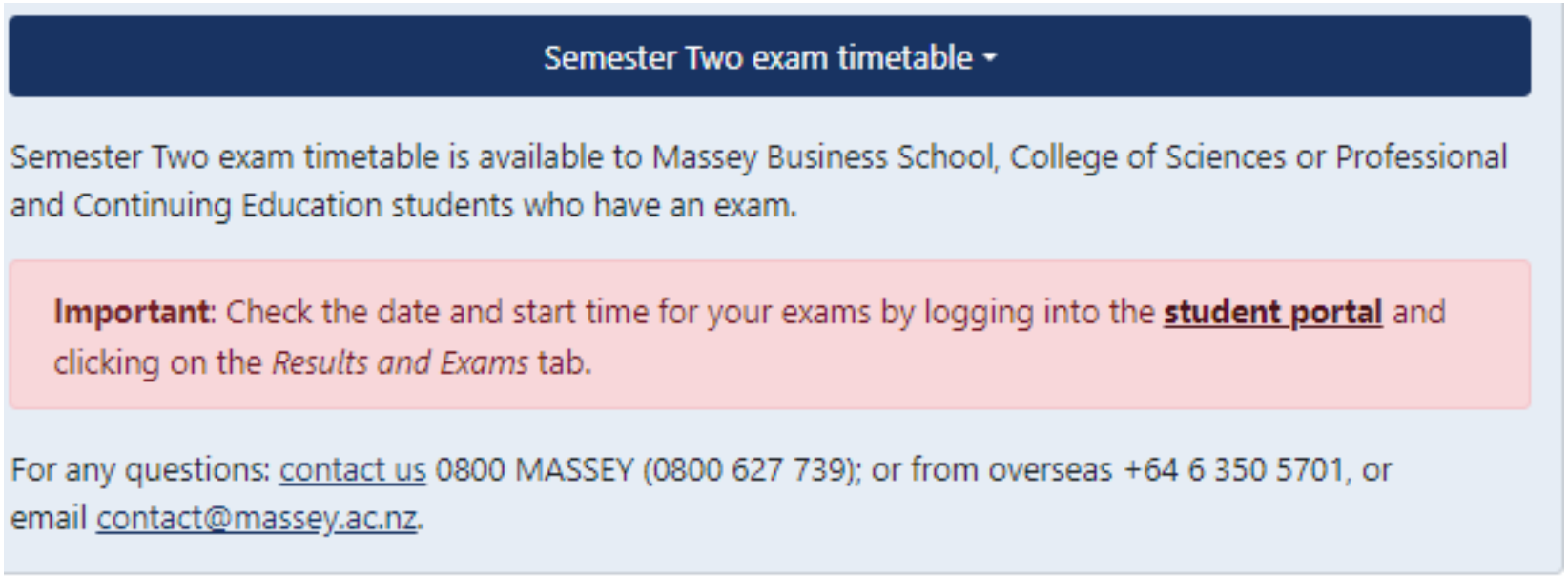
Online Exam

Syllabus



- Textbook - Chapters – 1, 2, 3, 4, 5, 6, 7, 8 and 14
- Lectures – Week 1 to 11
- Tutorial Exercises

Stream site


Semester Two exam timetable
Semester Two exam timetable is available to Massey Business School, College of Sciences or Professional and Continuing Education students who have an exam.
Important: Check the date and start time for your exams by logging into the student portal and clicking on the Results and Exams tab.
For any questions: contact us 0800 MASSEY (0800 627 739); or from overseas +64 6 350 5701, or email contact@massey.ac.nz.

Stream site

End-of-Semester Assessment – TCA (Time Constrained Assessment)

Everything you need for the Time Constrained Assessment (TCA) that will be your final assessment in this course will be available in this section:

- what is a TCA,
- what you need to do **BEFORE** your TCA,
- a link to helpful resources.

What is a Time Constrained Assessment (TCA)?

The TCA is an open book exam (i.e. you can consult your notes, textbooks etc), but students must not work together or ask any other person for help. T be in a Stream quiz, on this course Stream site.

You will access the questions at the scheduled date and time, for a set duration as noted in your exam timetable. We strongly encourage you to start y the scheduled time (***your LOCAL time***). The TCA must be completed within the 24-hour timeframe (based on the scheduled start time *in NZ time*).

Check the Student Portal - Examinations - Personal exam timetable to find the date and time of your TCA. If you are not able to sit on the scheduled da for an exam change via the Student Portal - Examinations tab.

This examination must be completed online. Your TCA should be completed on a computer or a suitable device where you can multi-task with Stream a applications effectively. Use of a mobile phone to complete the exam is **NOT** recommended. It is important to note, once you have started your TCA, y complete it within the time allocated. Stream will automatically close the quiz at the end of the set time.

There will be technical support available from 7am - 10pm NZST.

Although a TCA is an open book test, **you are not permitted to work together with other students, ask any other person for help, or offer any other person help.** The instructions explicitly forbid this. This also means you must not discuss the TCA on any platform for the entire 24 hours that the TCA is open.

Discussing the TCA, any questions, answers, or other details while it is still open gives other students unfair advantage and is a breach of academic integrity, and will be responded to accordingly.

Week 1

- Types of databases
- Why are databases valuable assets for decision making
- Importance of database design
- How modern databases have evolved from file systems
- Flaws in file system data management
- Main components of the database system
- Main functions of a DBMS

Week 2

- Data modeling
- Basic data-modeling building blocks
- Writing business rules
- Database design
- How data models have evolved
- Emerging alternative data models
- Classification of data models by their level of abstraction

Week 3

- Relational database model basic component
- Logical view of entities and relations
- Relational database operators
- Data dictionary
- The system catalog
- Data redundancy
- Indexing
- Relational algebra concepts – Unary and Binary

Week 4

- Entity Relationship Diagrams
- How entities and relationships are defined, refined, and incorporated into the database design process
- How ERD components affect database design and implementation
- Using flexible design to manage conflicting goals
- Characteristics of good primary keys

Week 5

- Bottom-up approach to database design
- Role of normalization in the database design process
- Improving database structure
- Normalization process - 1NF, 2NF, 3NF, BCNF, and 4NF
- Using normalization and ER modeling concurrently to produce a good database design
- Some situations which require denormalization

Week 6

- The basic commands and functions of SQL
- DDL – How to use SQL for data administration (to create tables and indexes)
- DML – How to use SQL for data manipulation (to add, modify, delete, and retrieve data)
- Queries – How to use SQL to query a database for useful information
  - CRUD (Create, Retrieve, Update, Delete) operations
  - Set operators UNION, UNION ALL, INTERSECT, MINUS
  - Virtual tables/ Views,
  - Virtual Columns

Week 7

Understanding stored procedures –

- PL/SQL,
- Functions,
- Scalar data types,
- Anchored data types
- Exception Handling.

Week 8

Understanding stored procedures –

- PL/SQL,
- Creating cursors and use them in stored procedures
- Writing triggers
- More Exception Handling.

Week 9

- LINQ - using the language syntax of C#
- Using LINQ with arrays
- Integrating Object-oriented (OO) concepts with LINQ syntax
- Using LINQ with table structures used in relational databases.
- SQL vs LINQ

Week 10

Big Data

- Evolution of data models
- Volume, Velocity and Variety
- Key Value Models
- NoSQL Databases
- MongoDB installation
- Queries in MongoDB
- NoSQL vs. SQL

Week 11

- Big Data
- MongoDB
- Parallel Processing
- Map Reduce & Aggregation
- SQL vs LINQ vs MapReduce/Aggregation

Review Questions

See screencast

All the best

